GREATER NOIDA INSTITUTE OF TECHNOLOGY
(ENGINEERING INSTITUTE)
Plot No-7, Knowledge Park-II, Greater Noida

News Shield: An Intelligent System for Detection and Verifying Fake News

PROJECT SYNOPSIS
OF MAJOR PROJECT

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Name of Student	Class Roll Number
Aryan Chourasia	2501320100070
Ayush kumar singh	2501320100080
Arpita Dixit	2501320100067
Ananya Singh	2501320100044

SUPERVISOR / GUIDE: Mr. Rohit Vaid

Greater Noida Institute of Technology, Greater Noida
Dr. A.P.J. Abdul Kalam Technical University, Lucknow

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NOTATIONS / NOMENCLATURE

Abbreviation	Technical Definition
ML	Machine Learning
NLP	Natural Language Processing
TF-IDF	Term Frequency - Inverse Document Frequency
SVM	Support Vector Machine
LSTM	Long Short-Term Memory
BERT	Bidirectional Encoder Representations from Transformers

1. INTRODUCTION

Background and problem statement

Digital communication allows news and claims to spread quickly through social media, websites, and messaging applications. Alongside legitimate reporting, fabricated, distorted, or misleading material may also be presented as news. Readers may have difficulty checking every item, while manual fact-checking takes time. The project addresses this need by proposing a quick screening tool for news-style text.

The proposed News Shield system will use supervised machine learning to classify a news article, headline, or social-media post into dataset-defined 'real' or 'fake' categories. It will learn patterns from labelled examples. The model output is an estimate based on those examples; it does not independently verify evidence, confirm the source, or establish that a claim is true.

Proposed approach and technology

The text-processing pipeline may include cleaning, tokenization, stop-word handling, and stemming or lemmatization. Candidate features include TF-IDF, Bag-of-Words, or word embeddings such as Word2Vec or GloVe. The proposed baseline comparison is Logistic Regression, Naive Bayes, Support Vector Machine, and Random Forest. If time and computing resources permit, LSTM/Bi-LSTM and BERT-based models may be explored as advanced comparisons.

The project proposes an interactive web application in which a user can enter a headline or article and receive a 'Real' or 'Fake' prediction with a model confidence score. Flask or Django or Streamlit may be used for the application, subject to the team's implementation choice. Any score will be labelled as a model estimate, not a verified probability of truth. The system should help readers decide what to check; important claims still require original and independent sources.

Field of project and special technical terms

The project belongs to the specialized field of Artificial Intelligence (AI) and Natural Language Processing (NLP). Key technical terms include:

- TF-IDF (Term Frequency-Inverse Document Frequency): A numerical statistic reflecting how important a word is to a document within a collection.
- Porter Stemmer: A heuristic morphological algorithm removing word suffixes to map tokens to their base form.
- Stylometry: The statistical measurement of writing style, including punctuation frequency, capitalization ratios, and sensationalist vocabulary.

Scope and expected contribution

The initial scope is text classification using a documented labelled dataset. Social-context signals, source-credibility indicators, transformer models, and deployment are possible extensions and will be included only if they can be implemented and evaluated within the project scope. Expected deliverables are a documented data pipeline, a model comparison using held-out data, an interactive prototype, and a report of errors and limitations. No accuracy level is assumed in advance.

2. LITERATURE REVIEW

Shu et al. survey fake-news detection from a data-mining perspective and organize prior work around news content, social context, datasets, and evaluation [1]. Their review highlights that text is only one possible signal. This supports the proposed text-focused baseline while motivating a clear statement that it does not use all available context.

Pérez-Rosas et al. introduced datasets covering multiple online news domains and evaluated linguistic approaches for automatic detection [2]. Ahmed, Traoré, and Saad studied n-gram analysis with machine-learning classifiers [3]. These works motivate testing conventional text features and classifiers, but their results belong to their specific datasets and experimental settings; they do not guarantee the same performance for this project.

Wang's LIAR benchmark contains manually labelled short political statements and provides a benchmark for fine-grained classification [4]. Ruchansky et al.'s CSI model combines article text with user-response and source-behaviour information [5]. These works show the difference between a text-only prototype and systems that incorporate context or social signals.

Kaliyar et al. proposed FakeBERT, combining BERT representations with convolutional layers for social-media fake-news classification [6]. Devlin et al. introduced BERT as a bidirectional language representation model [8]. FakeNewsNet provides news content together with social and spatiotemporal information for research [7]. The team's supplied draft also identifies the Kaggle Fake and Real News dataset and PolitiFact as possible data or fact-checking resources [9], [10]. Availability, licensing, labels, and suitability must be checked before choosing a dataset.

The reviewed work ranges from conventional n-gram models to contextual deep-learning and social-context methods. The proposed mini/major project will first establish a reproducible baseline, then compare selected alternatives if resources permit. Evaluation on held-out examples, error analysis, and human verification are necessary because a classifier can fail on new domains, satire, or changing language.

3. OBJECTIVES

Based upon the literature review and the identified research problem, the primary goal of News Shield is to design, evaluate, and benchmark a reliable machine learning system for news credibility assessment. The general objective is broken down into eight logically connected parts:

1. Literature & State-of-the-Art Review: Study existing fake news detection approaches, benchmark datasets (LIAR, FakeNewsNet, ISOT), feature representations, and evaluation frameworks relevant to the selected project scope.

2. Dataset Selection & Curation: Select, balance, and document a labelled real/fake news dataset across diverse domains (politics, science, health, technology), recording provenance, license, and known limitations.

3. Text Preprocessing & Cleaning: Implement a standardized NLP pipeline including URL/HTML stripping, non-alphabetic filtering, stop-word elimination, and morphological stemming (Porter Stemmer).

4. Feature Extraction & Representation: Compare TF-IDF (unigrams and bigrams with sublinear scaling) and Bag-of-Words representations to extract discriminating lexical and stylometric patterns.

5. Baseline Classifier Benchmarking: Train and evaluate multiple supervised classification models, including Multinomial Naive Bayes, Logistic Regression, Support Vector Machines (Linear SVM), and Random Forest.

6. Rigorous Empirical Evaluation: Measure generalization performance using 5-Fold Stratified Cross-Validation, test accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix analysis on held-out data.

7. Interactive Prototype Development: Build a responsive Streamlit web application providing real-time article verification, model confidence meters, stylometric indicators, and batch CSV processing.

8. Error Analysis & Limitation Reporting: Analyze false-positive and false-negative boundaries, model uncertainty, and domain-transfer constraints to ensure responsible deployment as a screening aid.

4. METHODOLOGY / PLANNING OF WORK

The project methodology comprises eight systematic development stages executed sequentially during project completion to ensure rigorous engineering and scientific reproducibility:

Stage 1: Requirements Formulation & Scope Definition: Define the problem boundaries, project scope, dataset permissions, evaluation criteria, and user interface requirements in consultation with the project guide.

Stage 2: Dataset Acquisition & Exploratory Analysis: Acquire and verify a balanced corpus of real and fake news articles; inspect class distributions, missing values, text-length variations, and potential duplicate samples.

Stage 3: Data Partitioning & Preprocessing Pipeline: Partition the dataset into stratified 80% training and 20% test subsets. Fit preprocessing transformations (URL stripping, tokenization, stop-word filtering, stemming) strictly on training data.

Stage 4: Feature Engineering & Vector Space Construction: Construct high-dimensional numerical feature matrices using TF-IDF vectorization (unigrams + bigrams, 8,000 max features, sublinear TF scaling) and stylometric indicators.

Stage 5: Classifier Training & Hyperparameter Tuning: Train candidate baseline classifiers (Multinomial Naive Bayes, Logistic Regression, Linear SVM with Platt probability calibration, Random Forest, and Ensemble Voting).

Stage 6: Multi-Metric Model Validation & Benchmarking: Evaluate models using 5-Fold Stratified Cross-Validation on training data and hold-out testing for Accuracy, Precision, Recall, F1-Score, and Confusion Matrix generation.

Stage 7: Web Application Interface Implementation: Develop an interactive web dashboard (Streamlit) supporting real-time headline/article input, confidence meters, batch CSV analysis, and comparative benchmark visualizations.

Stage 8: Robustness Testing & Final Documentation: Perform stress testing on challenging unseen news articles, conduct error boundary analysis, document operational limits, and compile project documentation.

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